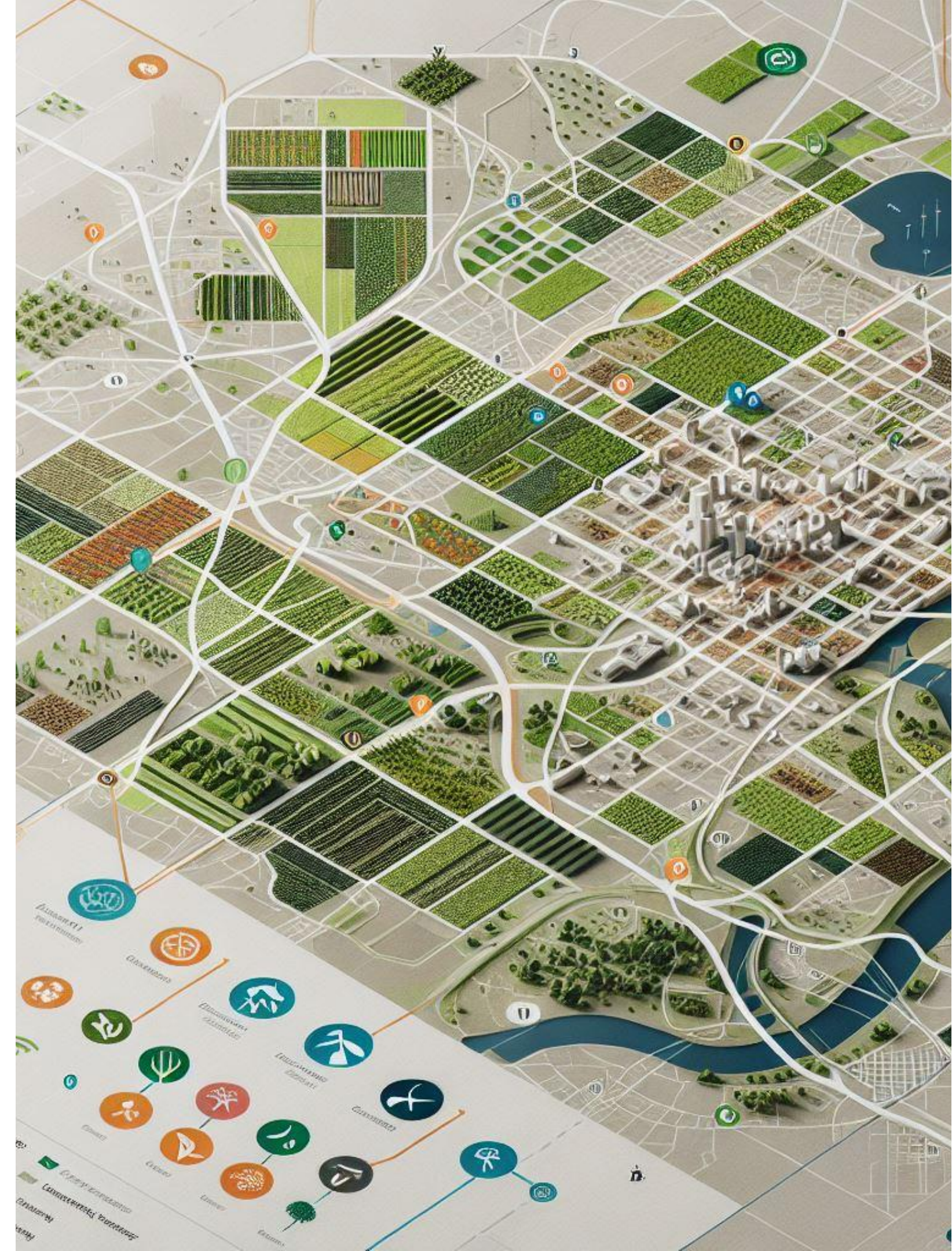


Geospatial modelling for sustainable urban food farming systems. Example of the metropolis of Vilnius, Lithuania.

AIX-MARSEILLE-UNIVERSITE
GEOMATIQUE ET MODELISATION SPATIALE

HGAU39 Remote sensing

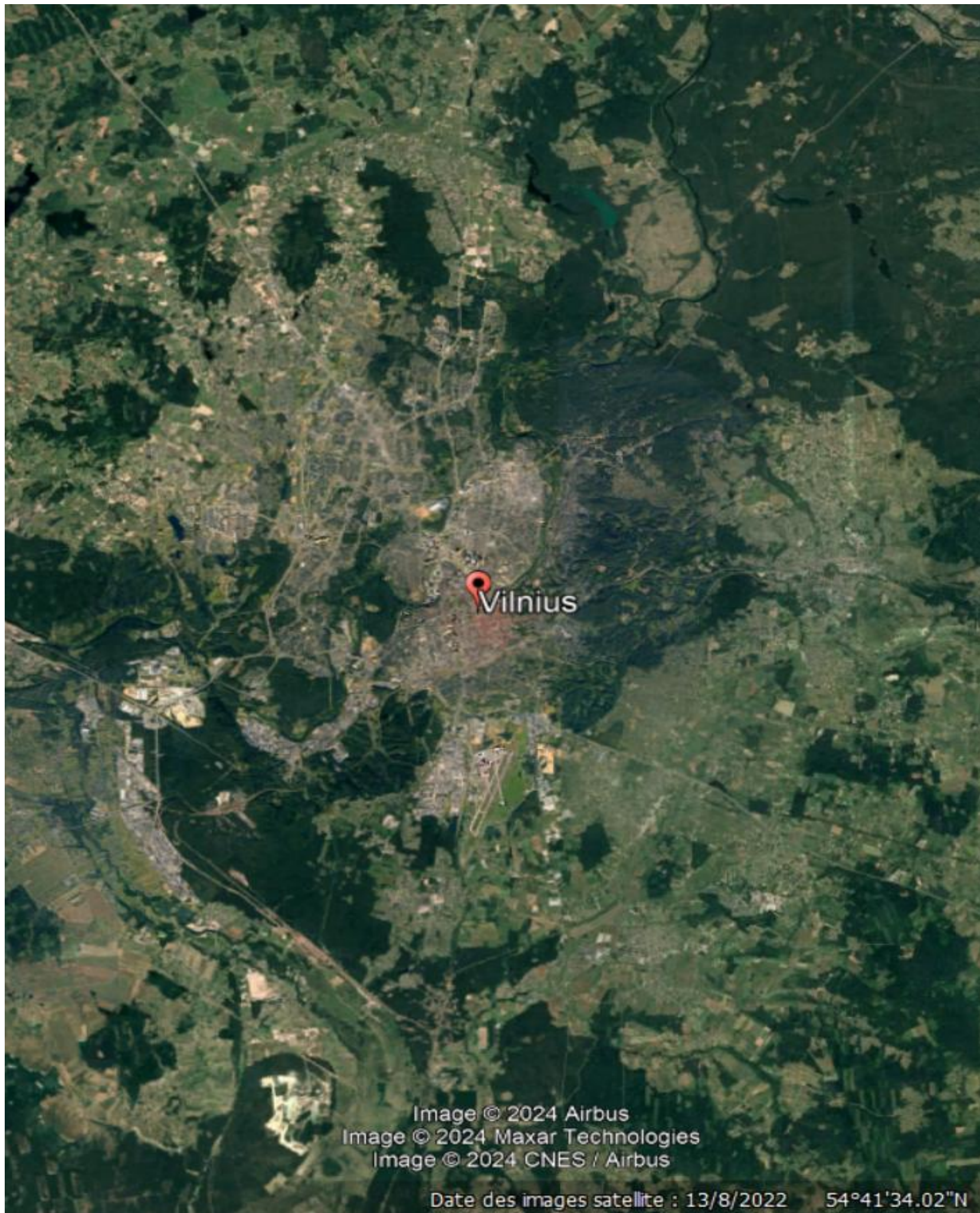




I. INTRODUCTION

CONTEXTUALIZATION

- Rapid growth of urban populations.
- Pressures on natural resources and infrastructure.
- Need for innovative solutions for food security.
- Food production in urban areas.
- Reduction of the carbon footprint related to the transport of food.
- Contribution to the resilience and sustainability of cities.
- Optimization of the use of urban spaces for agriculture.



Problematic

How can geospatial modelling tools be used to identify the most suitable urban areas for sustainable urban agriculture in Vilnius?

Presentation of the study area

Geographically

- Vilnius, the capital of Lithuania, is located in the southeast of the country, about 250 km from the Baltic Sea;
- The city covers an area of 401 km² and is crossed by the Neris and Vilnia rivers, which contribute to its picturesque landscape;
- The terrain is varied, with hills and valleys, including Gediminas Hill, which offers panoramic views of the city;
- Vilnius' climate is continental, with cold winters and moderate summers.

Demographics

- Vilnius is the largest city in Lithuania, with a population of around 581,000 inhabitants in 2023;
- The population density is about 1,450 inhabitants per km²;
- The city is culturally diverse, with a population that includes Lithuanians, Poles, Russians, and other ethnic groups, which contributes to its cosmopolitan atmosphere.

In terms of agriculture

- Although Vilnius is primarily an urban city, agriculture plays an important role in the surrounding area;
- The area around Vilnius is characterized by fertile agricultural land, where cereals, vegetables and fruits are mainly grown.

Geographical objects to study

Category	Description	Indicators	Analytical methods
Town	Vilnius, capital of Lithuania	Urban population, population density	Analyse démographique, SIG (Système d'Information Géographique)
Residential areas	Varied urban neighbourhoods with different population densities	Residential density, types of dwellings	Cartography, spatial analysis
Green spaces	Parks, public gardens, and other open spaces	Surface area of green spaces, accessibility	Remote sensing, land use analysis
Infrastructure	Transportation networks, commercial and industrial buildings	Infrastructure distribution, accessibility	GIS, network analysis

Soil Analysis	Types of soils and their suitability for urban agriculture	Soil composition, fertility	Remote sensing
Climatic Factors	Local climatic variations affecting agricultural production	Temperature, humidity, precipitation	Remote sensing, climate modeling, meteorological data analysis
Access to Water	Availability and management of water resources for irrigation	Water availability, water quality	Hydrological analysis, water resource mapping (remote sensing)
Existing Infrastructure	Use of current infrastructure to support urban agriculture	Infrastructure capacity, adaptability	Infrastructure usage analysis, GIS
Food Resilience	City's ability to maintain a sustainable food supply	Food security, crop diversity, agricultural yield	Resilience modeling, food systems analysis

II. Data and methods

1.Data presentation

1.1. Landsat 8

Bands	Wavelength (μm)	Resolution (m)	Characteristics	Applications
Bands 2	0.450 - 0.515	30	Visible blue, useful for soil and water analysis	Soil mapping, distinction between vegetation
Bands 3	0.525 - 0.600	30	Visible green, accentuates vegetation	Plant vigour assessment
Bands 4	0.630 - 0.680	30	Visible red, discriminating vegetation	Crop Health Analysis
Bands 5	0.845 - 0.885	30	Near Infrared (NIR), sensitive to plant moisture	Biomass estimation, calculation of the NDVI index
Bands 6	1.560 - 1.660	30	Shortwave Infrared (SWIR), useful for soil moisture	Soil moisture analysis
Bands 7	2.100 - 2.300	30	Shortwave Infrared (SWIR), discriminating between soil types	Geological studies, distinction between dry and wet soils

1.2. Sentinel 2

Bands	Type	Resolution	Wavelength (nm))	Applications
B2	Bleue	10 m	490	Water and vegetation quality analysis
B3	green	10 m	560	Vegetation Health Assessment
B4	Red	10 m	665	Soil mapping and vegetation analysis
B5	Red Edge (Visible and NIR)	10 m	705	Crop health monitoring
B6	Near-infrared	10 m	740	Soil and vegetation moisture analysis
B7	Near-infrared	10 m	783	Plant and forest health monitoring
B8	Near-infrared	10 m	842	Vegetation analysis and crop detection

1.3. Indices and indexes

Indice/Index	Formula	Characteristics	Applications
Normalized Difference Vegetation Index (NDVI)	$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$	Measuring plant health and vigour	Monitoring crop growth, assessing ecosystem health
Normalized Difference Water Index (NDWI)	$\text{NDWI} = (\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$	Water Presence Indicator	Water resources monitoring, wetland management
Enhanced Vegetation Index (EVI)	$\text{EVI} = 2.5 * (\text{NIR} - \text{Red}) / (\text{NIR} + 6 * \text{Red} - 7.5 * \text{Blue} + 1)$	Sensitive to variations in vegetation, less affected by shade	Monitoring of changes in vegetation cover, assessment of productivity

SCHEMA METHODOLOGIQUE

Geospatial modelling for sustainable urban food farming systems. Example of the metropolis of Vilnius, Lithuania.

Download Landsat 8 imagery
on Earth Explorer from 2015
and 2024

QGIS

Image Processing on
Terrset

B3/B5

NDWI

B4/B5

NDVI

B5/B4/B2

EVI

B4/B8

B1,B2,B3,
B4,B5,B6,
B7

Cluster

B2,B3,B4,
B5,B6,B7,
B8

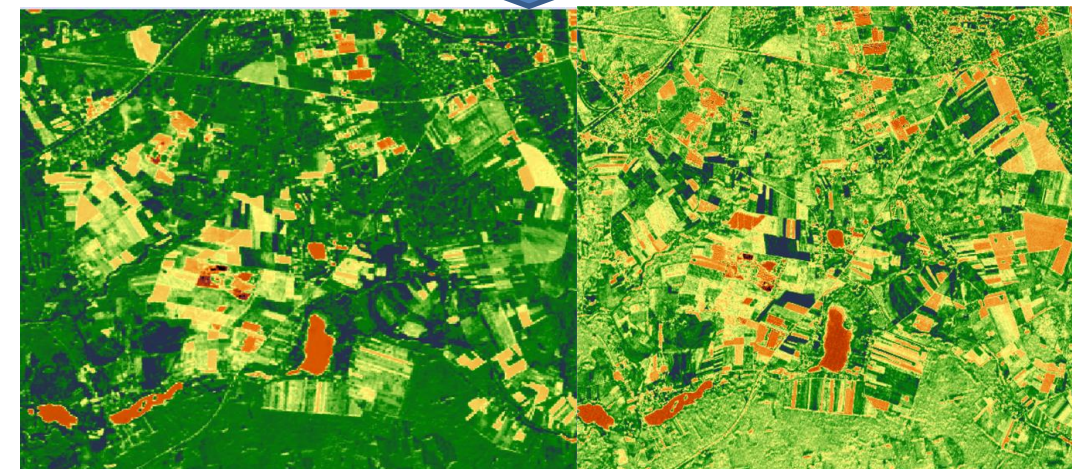
Download the 2024 Sentinel 2
images on Copernicus

QGIS

Image Processing on Terrset

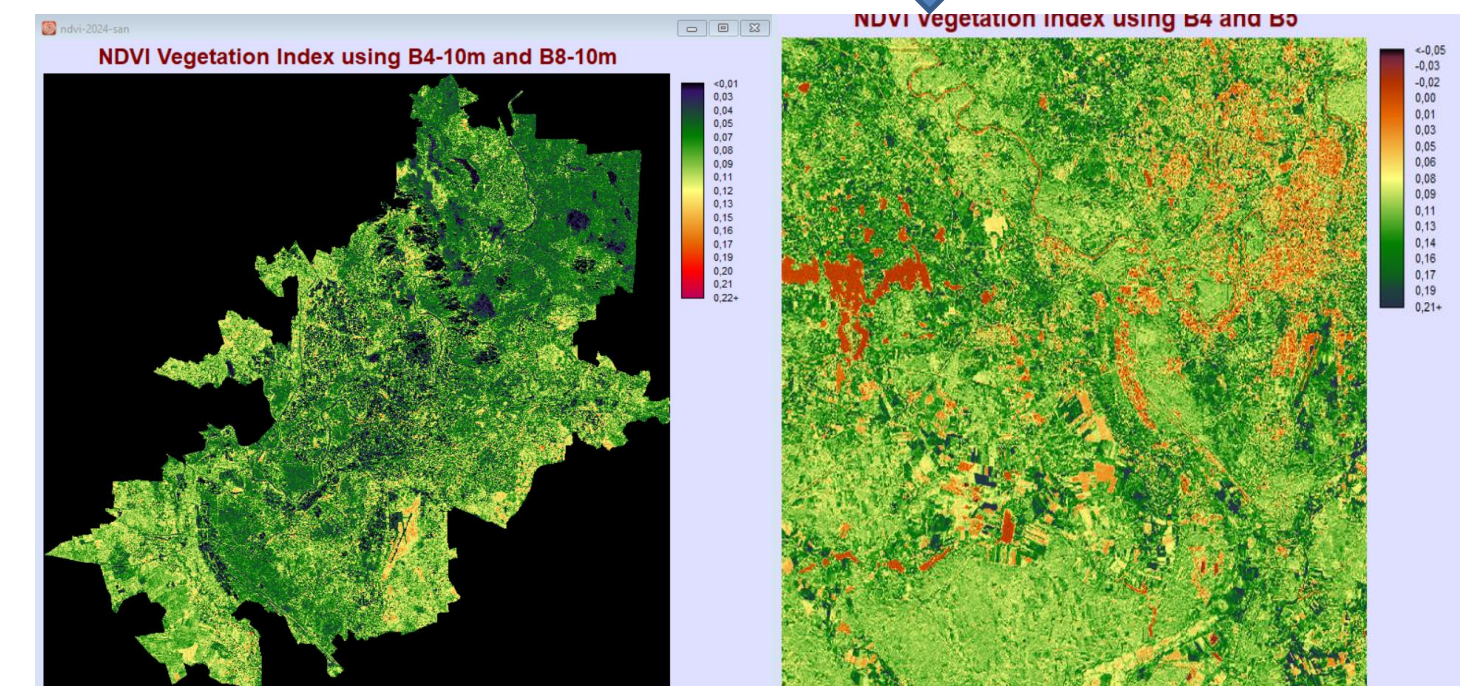
Results

Comparison of Landsat 2024 and
Sentinel 2024 imagery



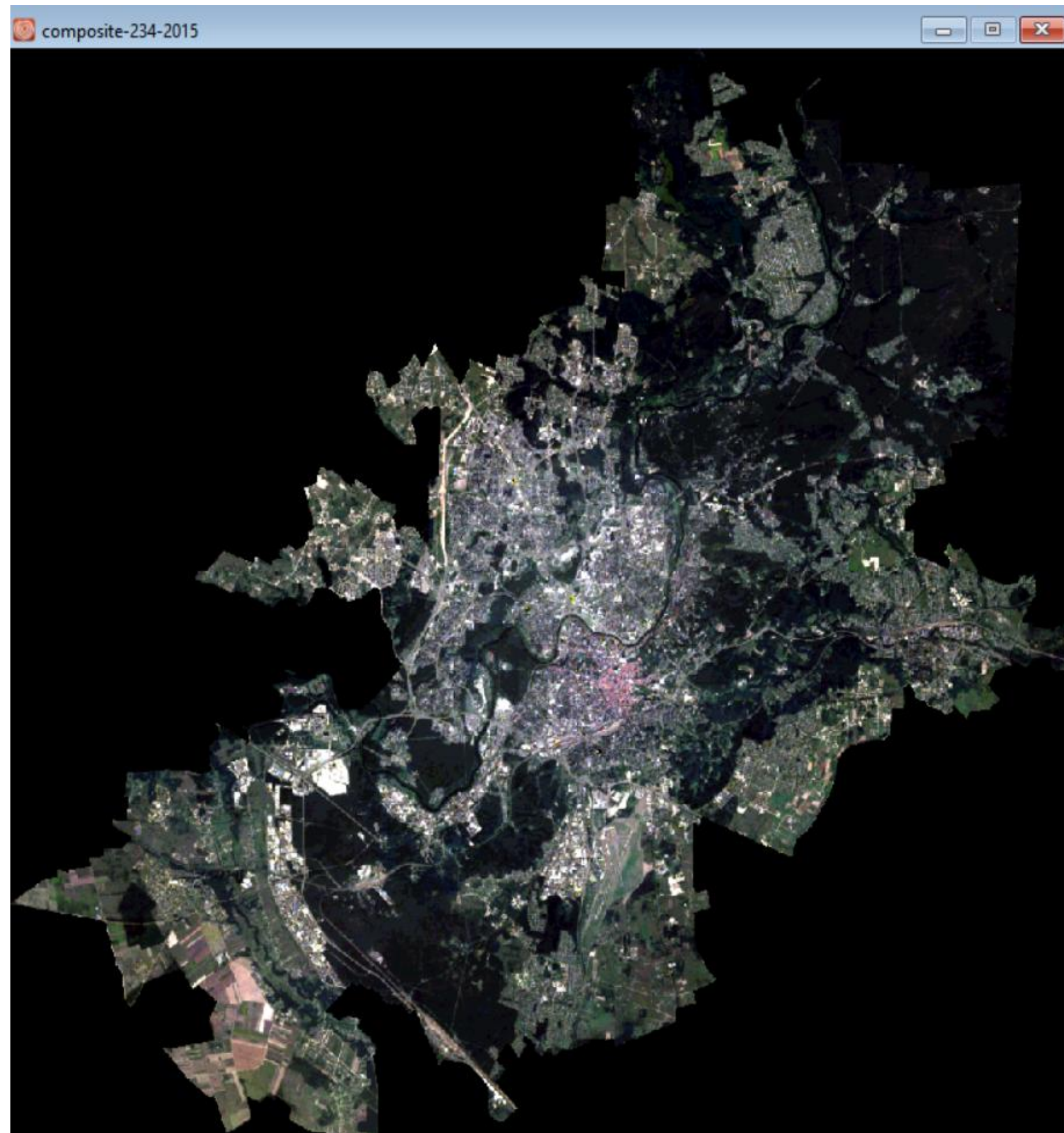
Landsat 2024

Sentinel 2



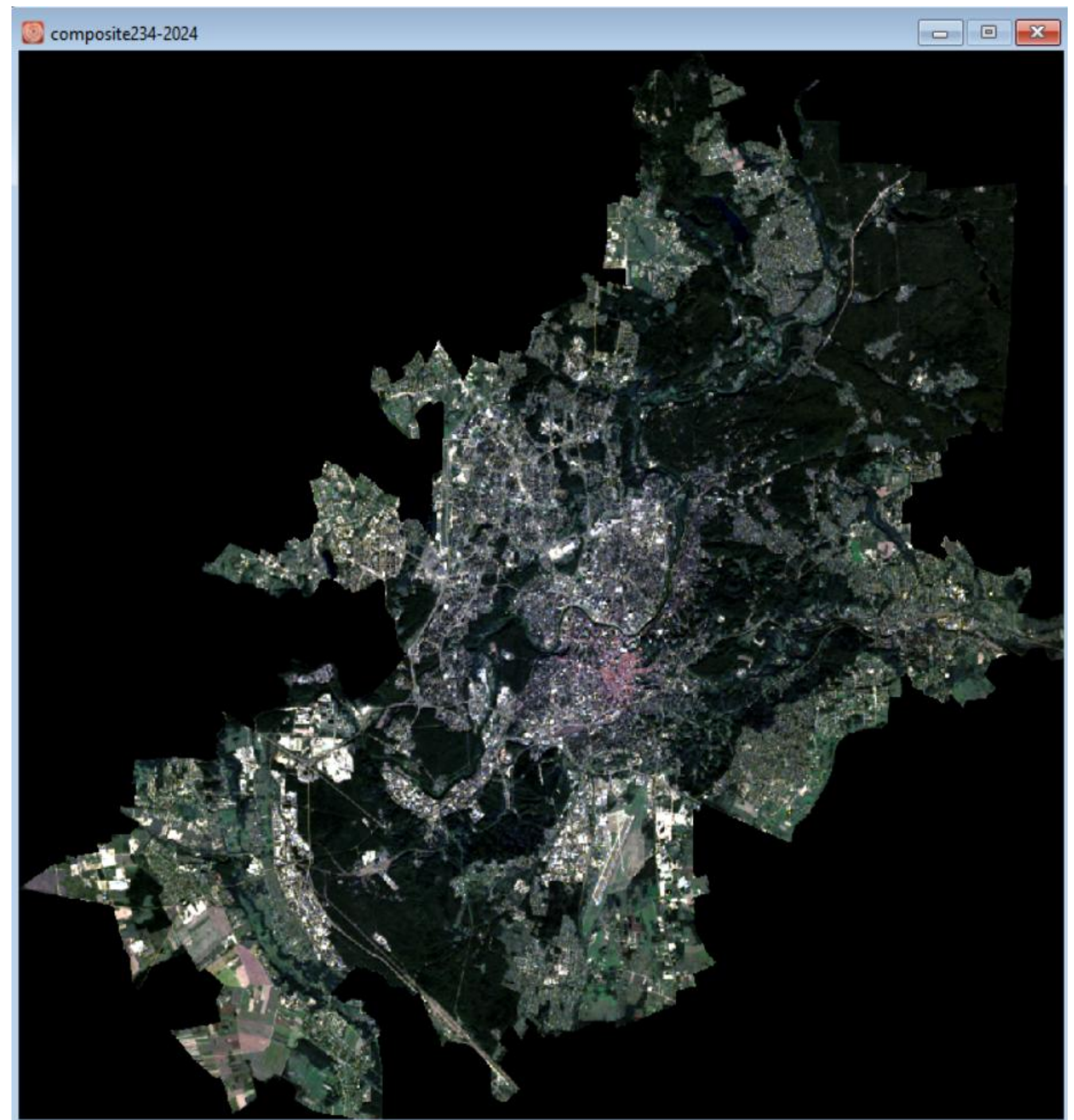
TREATMENT OUTCOMES

Composite-2015



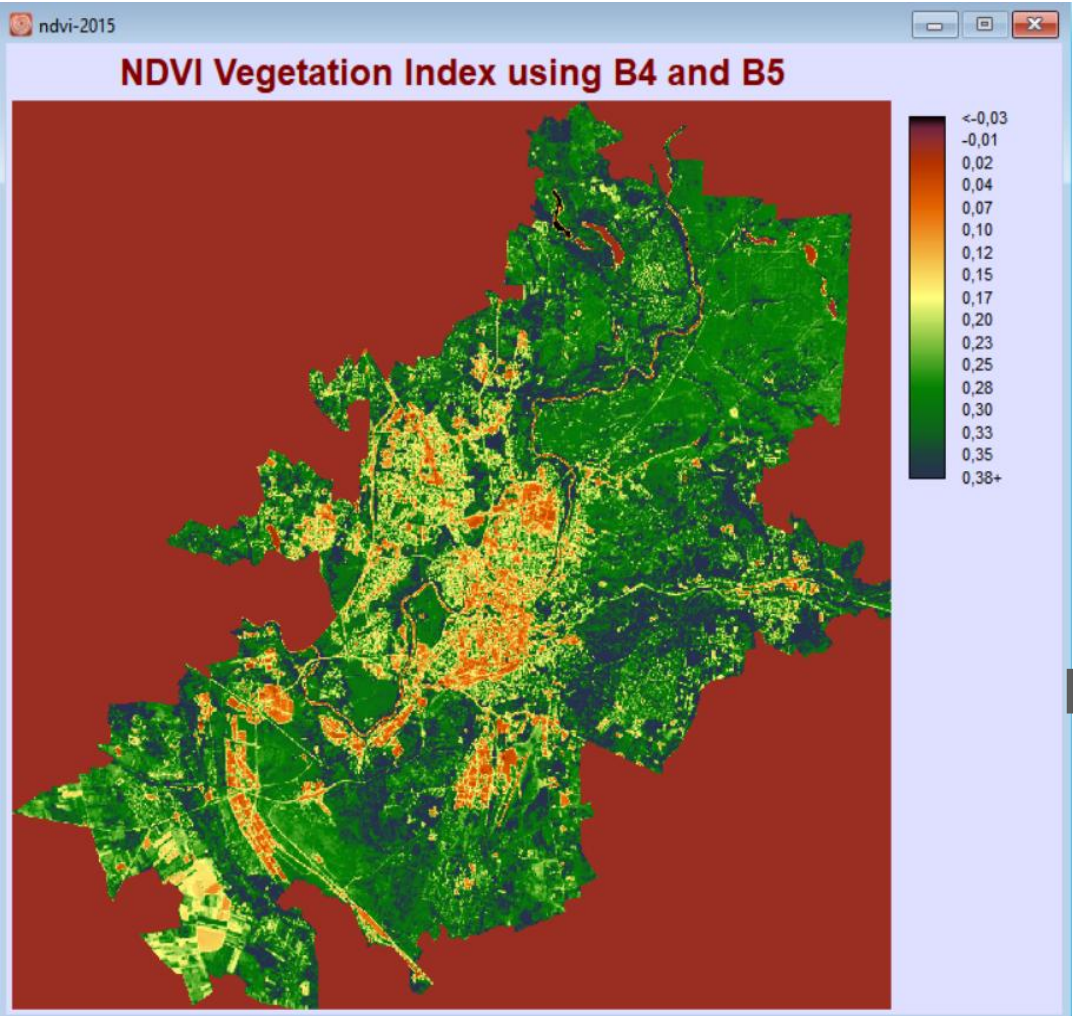
Source: Landsat-8-b2,b3,b4

Composite-2024

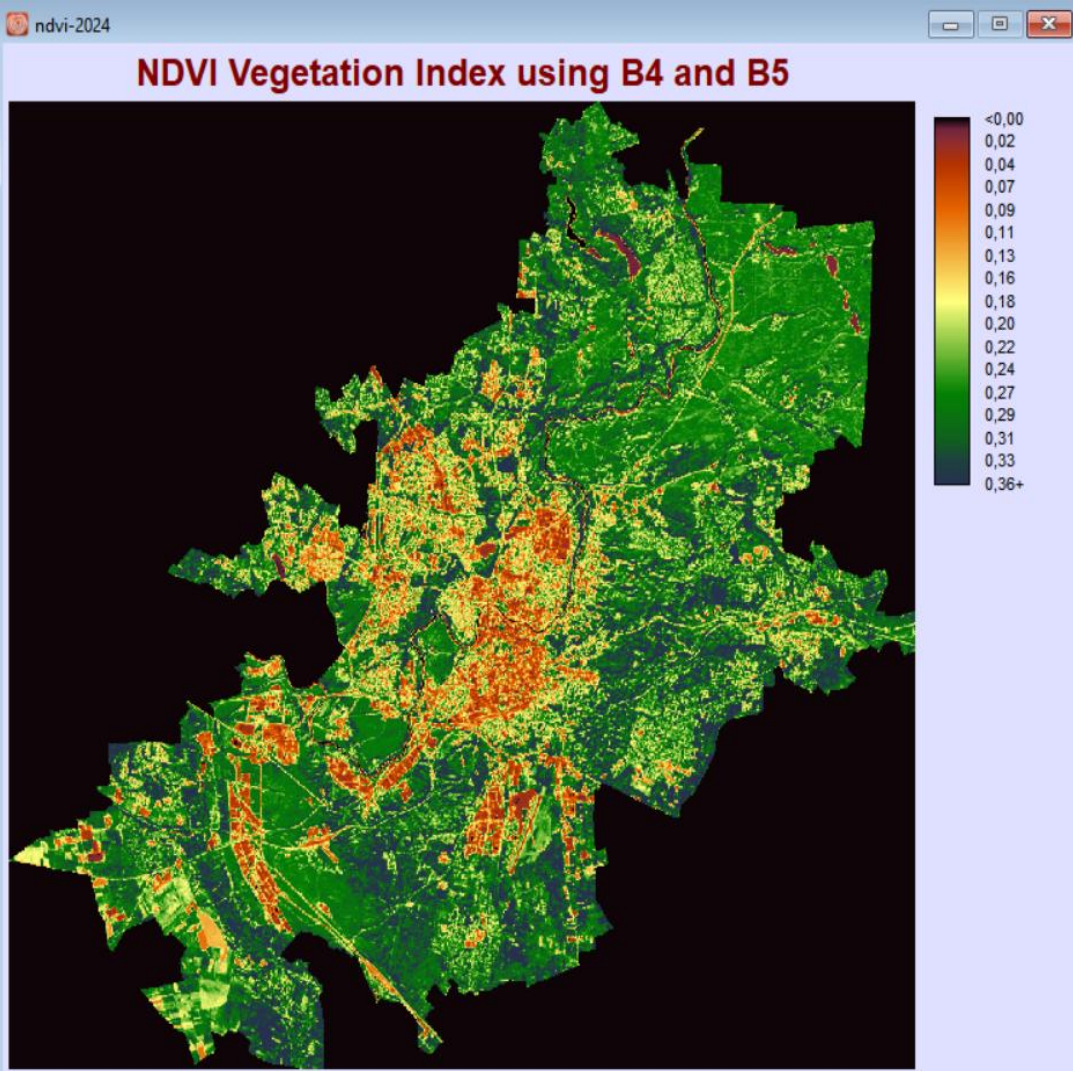


Source: Landsat-8-b2,b3,b4

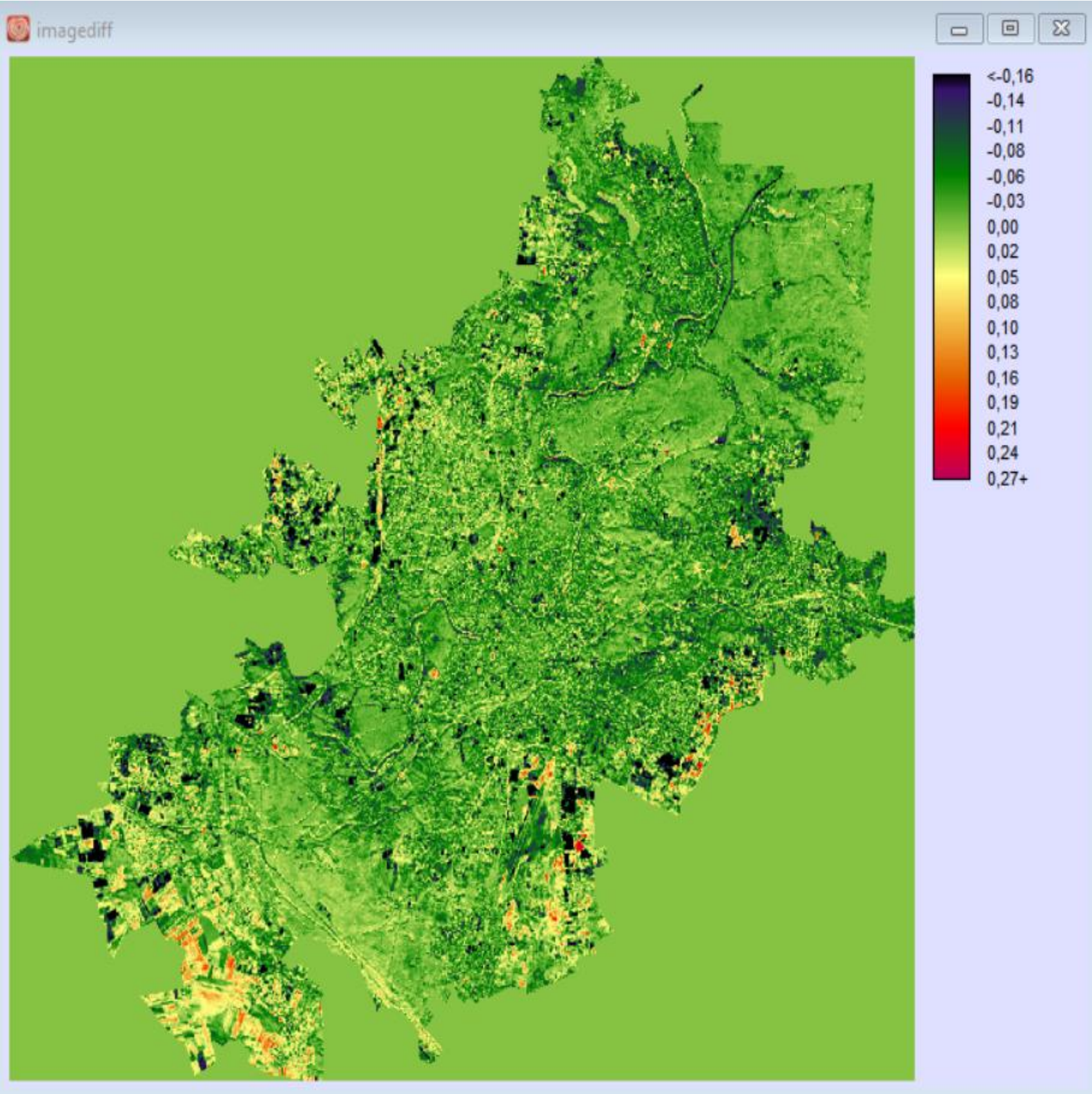
NDVI-2015



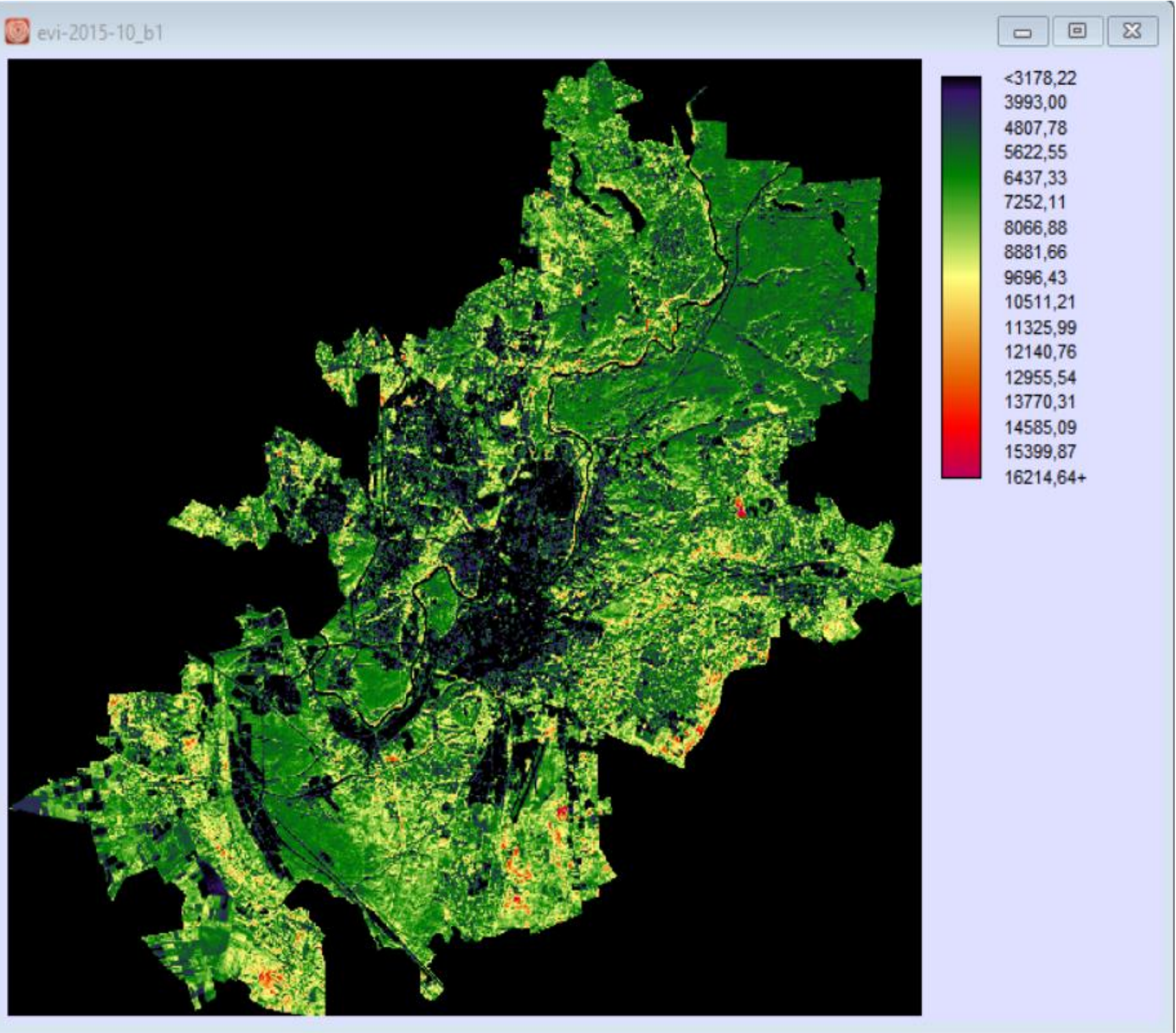
NDVI-2024



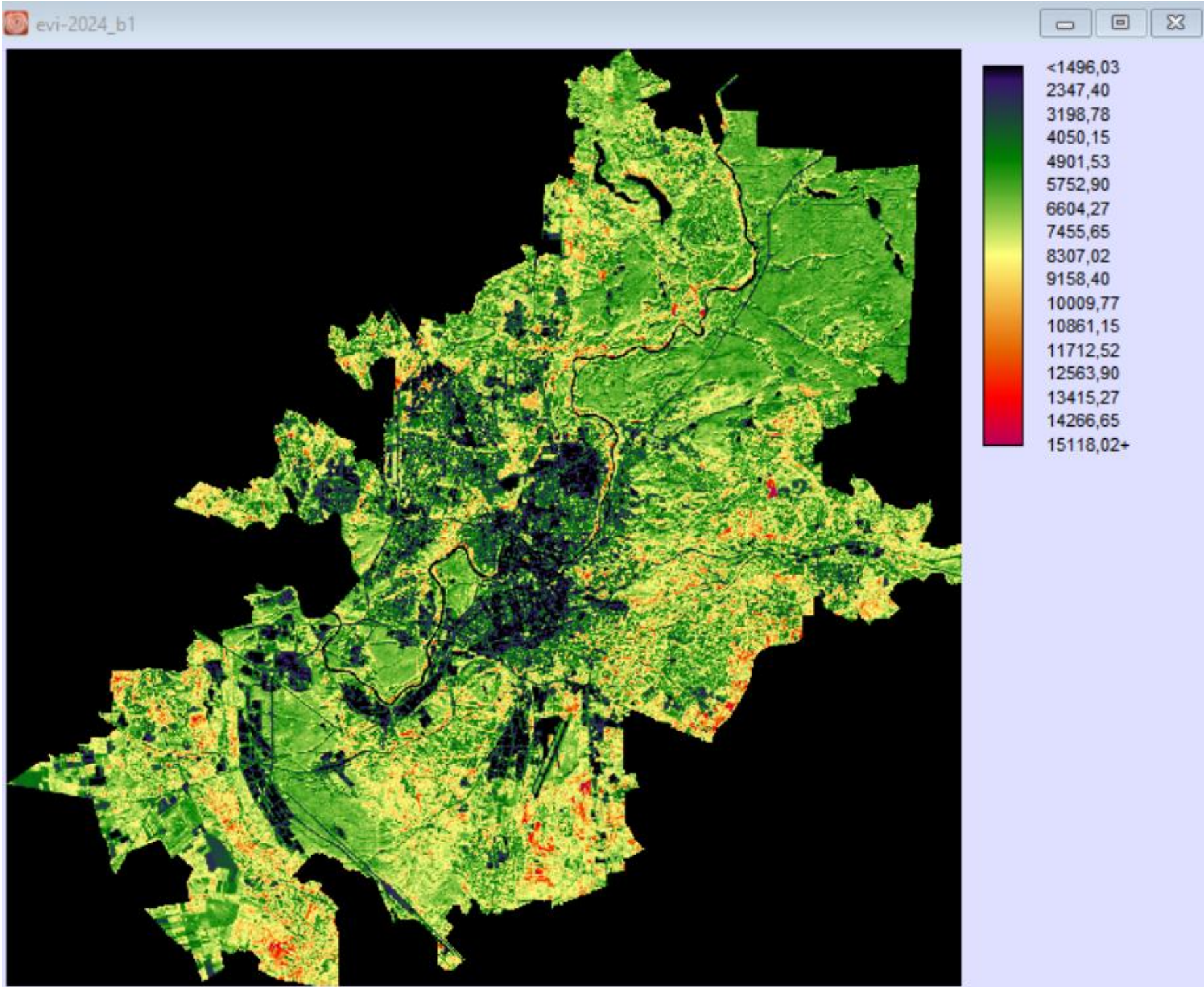
IMAGEDIFF NDVI 2015-NDVI 2024

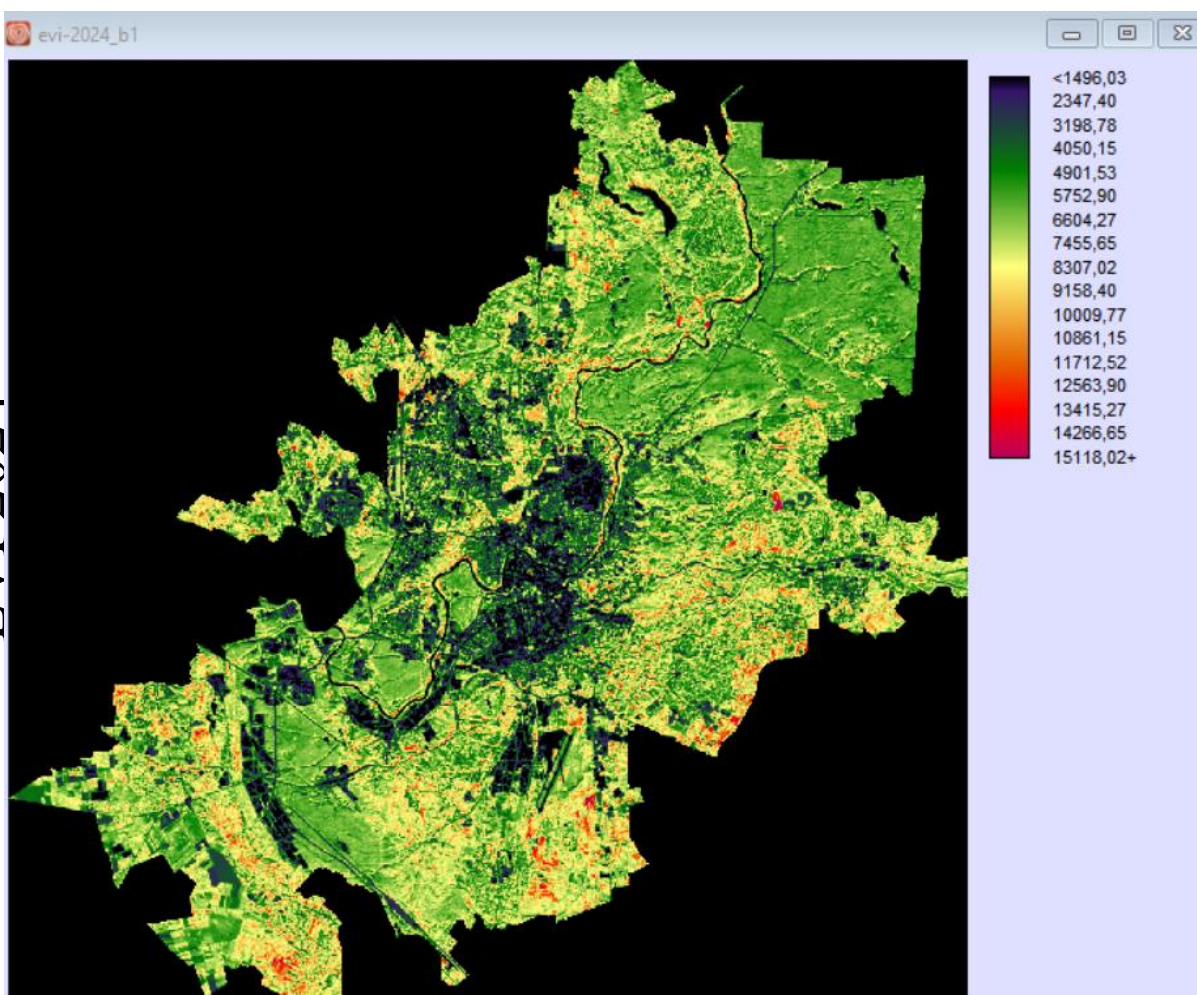
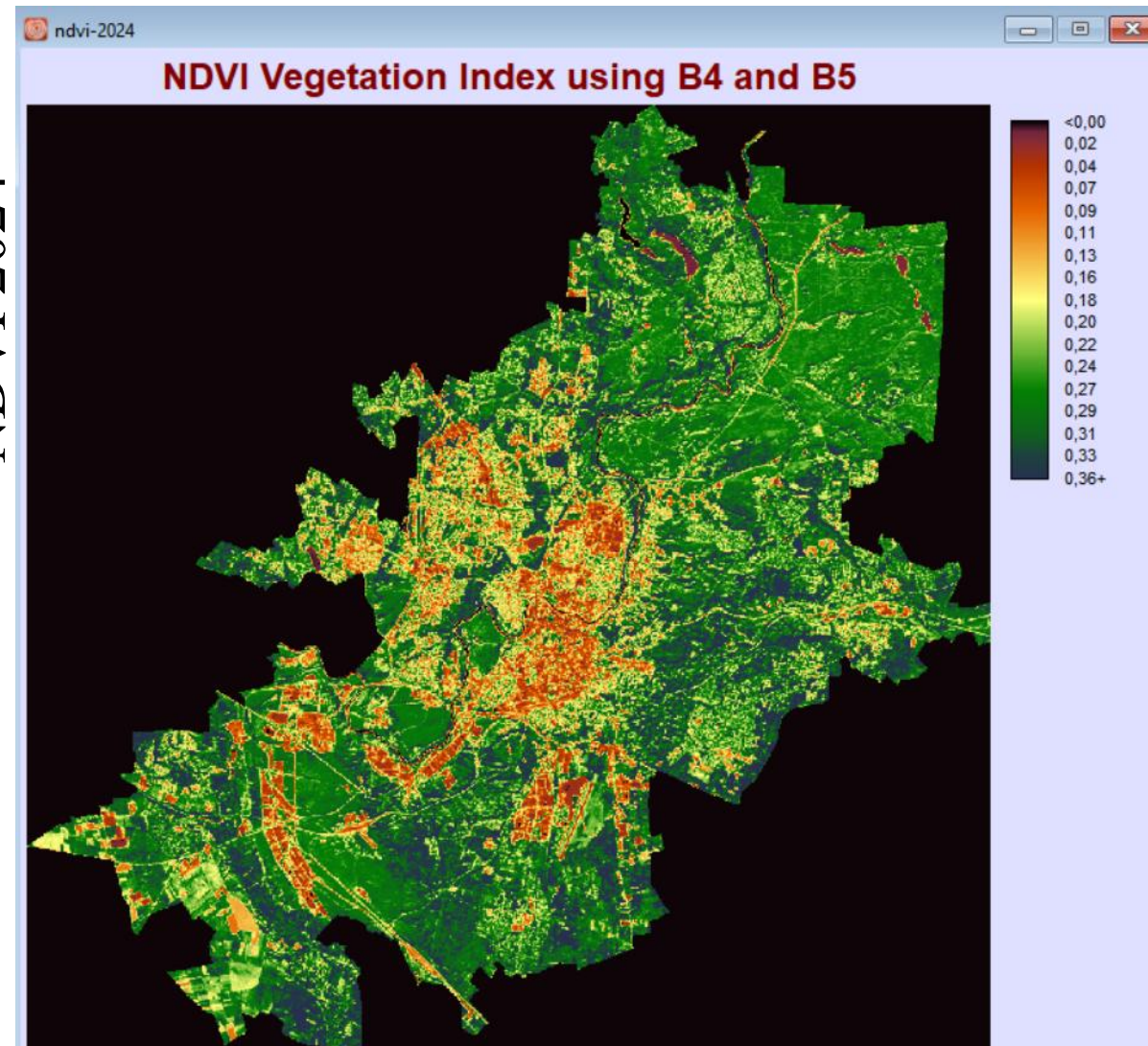


EVI-2015

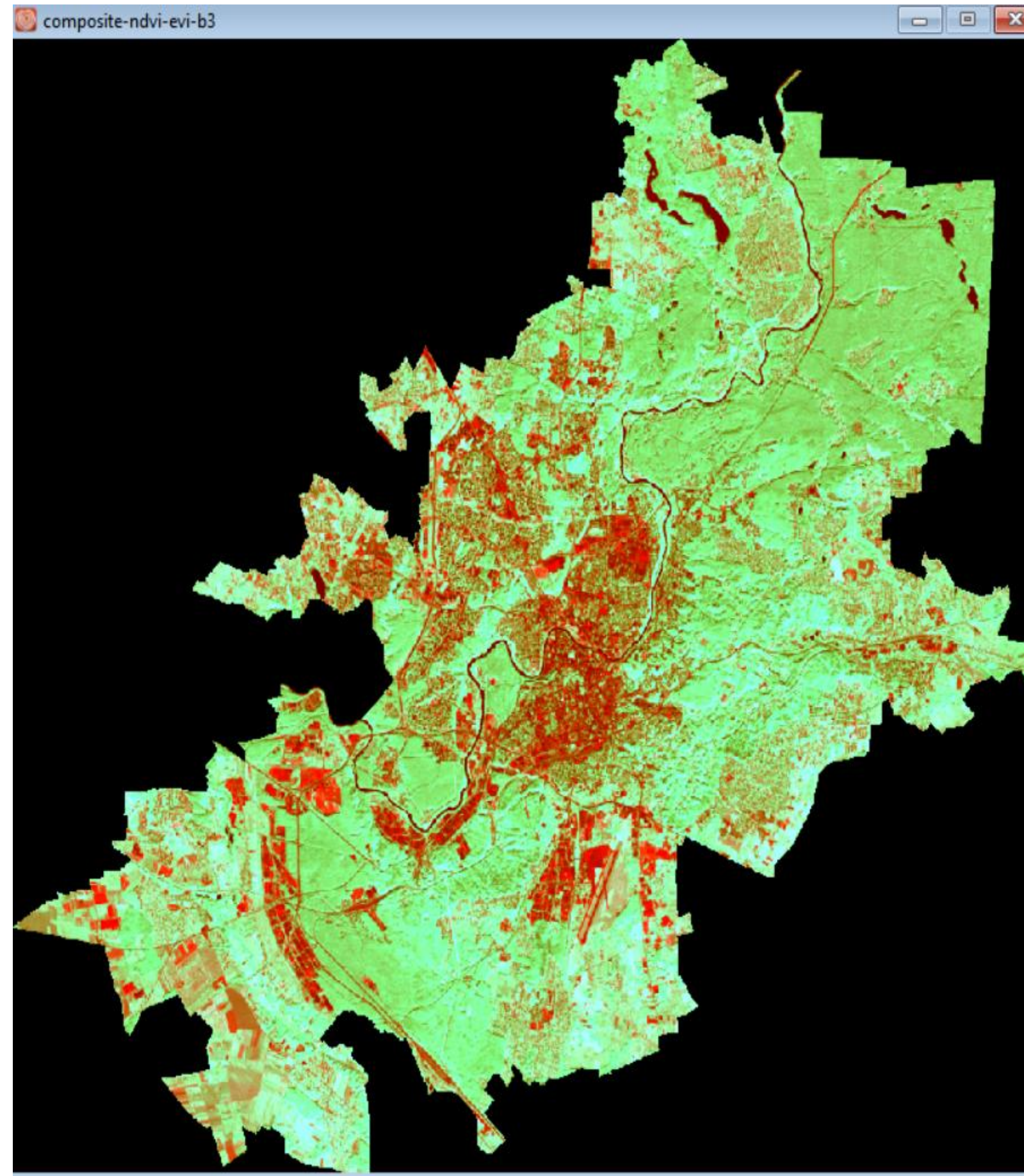


EVI-2024





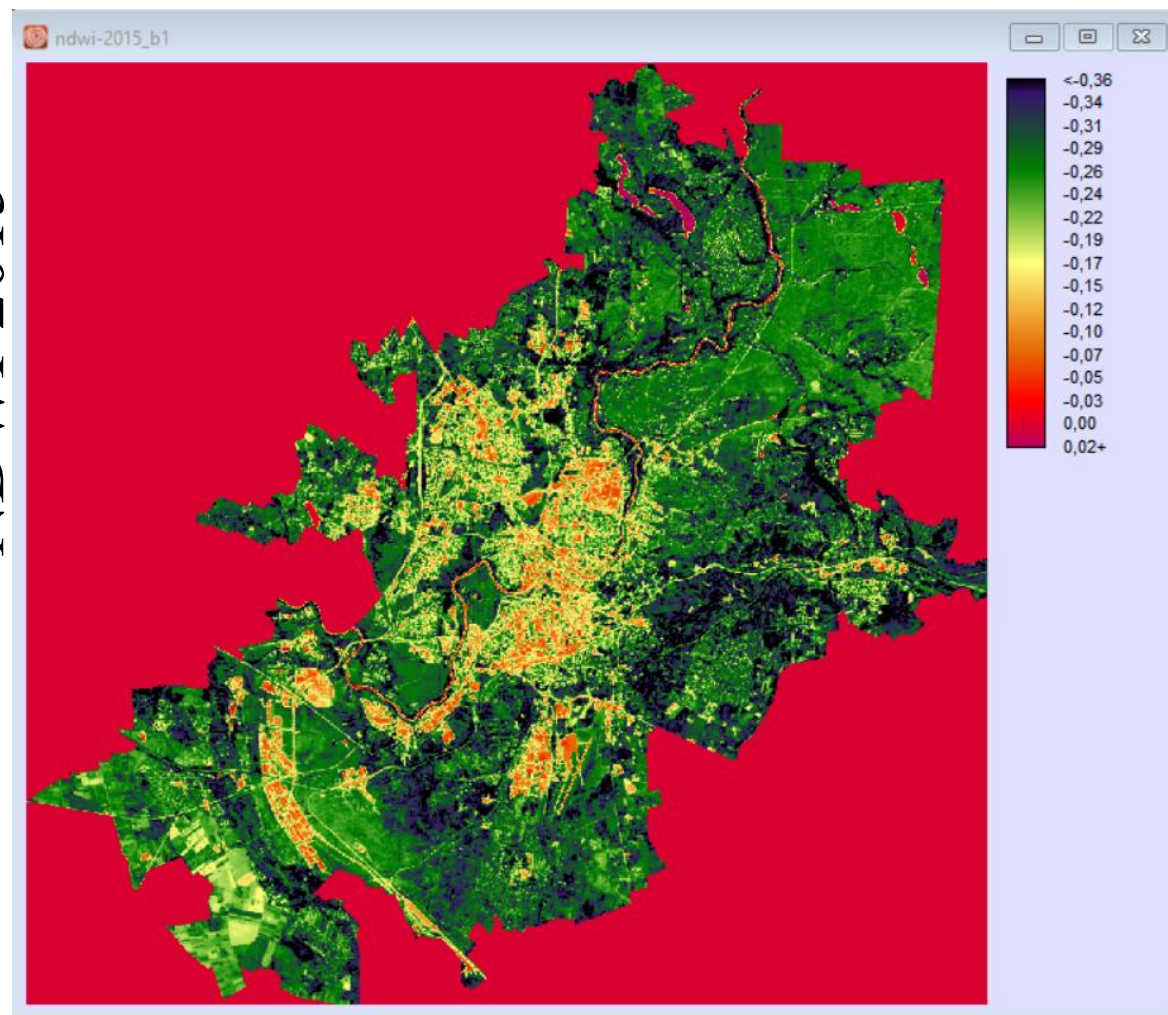
Composite NDVI-EVI-B3



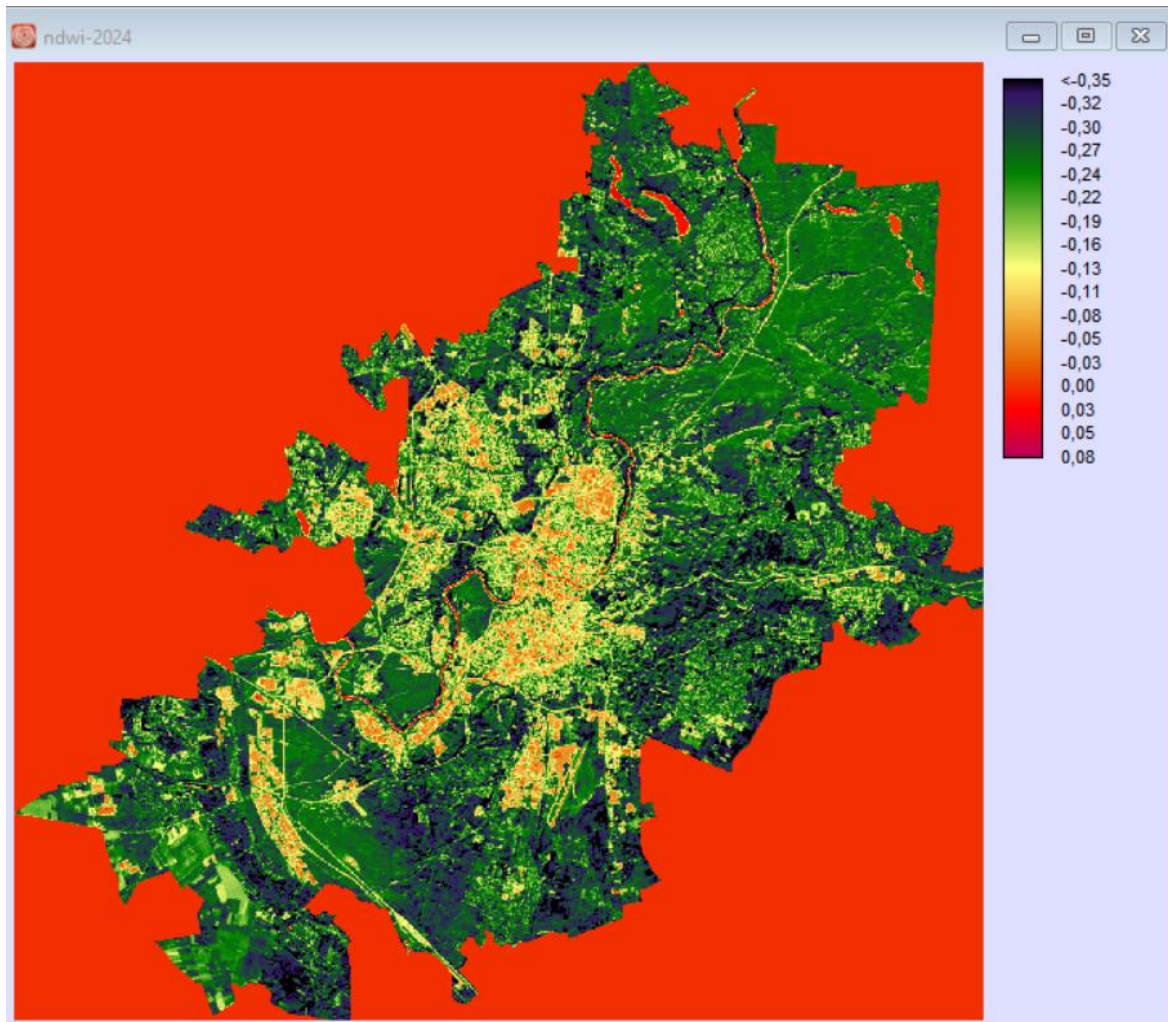
EVI is often considered an improvement in NDVI, especially under conditions of dense vegetation or when atmospheric corrections are required.

The combination of the two indices (NDVI, EVI) allowed us to clearly see the areas where the vegetation is very dense. For example, areas that have a low presence of red color, show us that the vegetation is healthy and very dense.

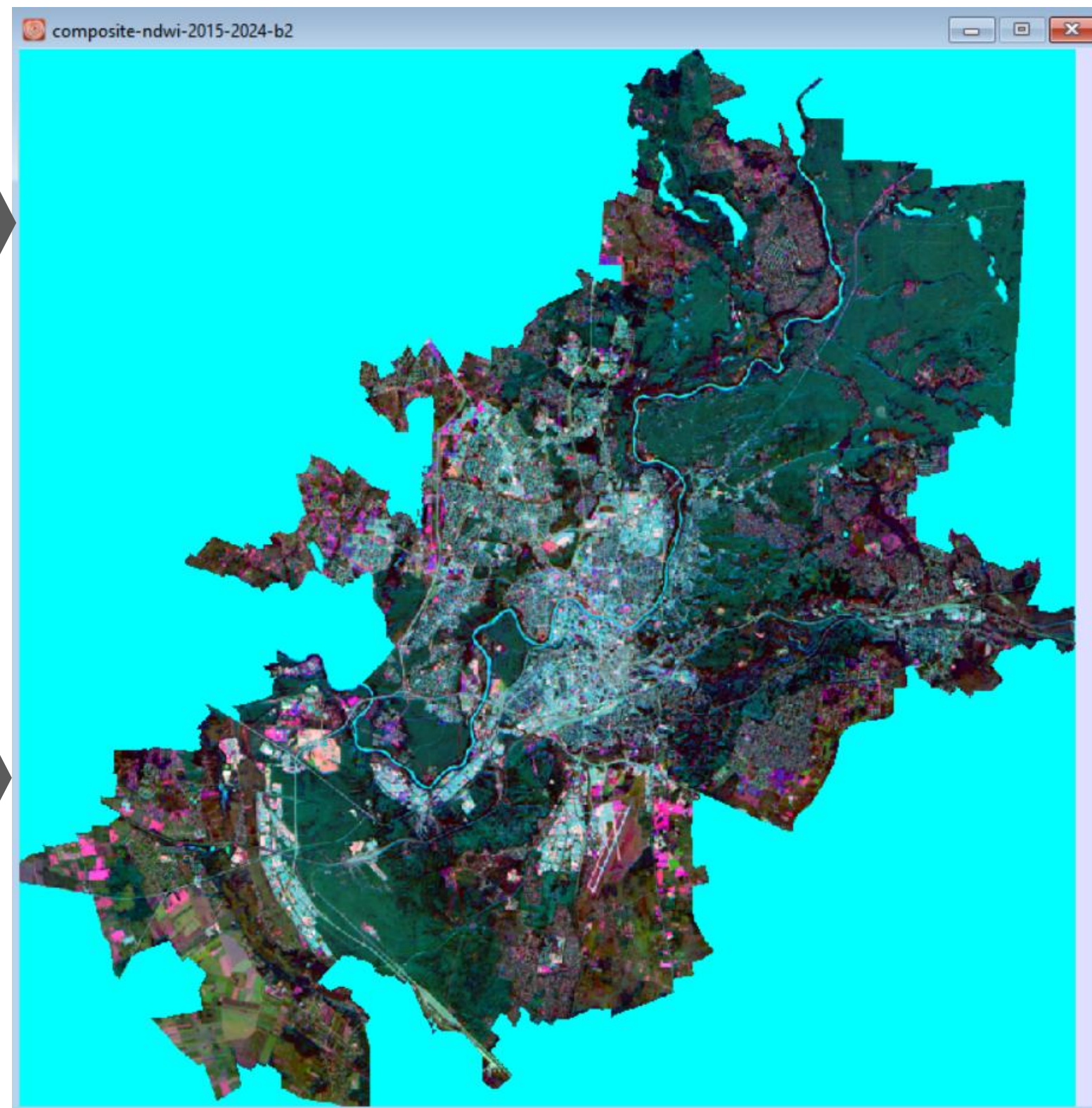
NDWI-2015



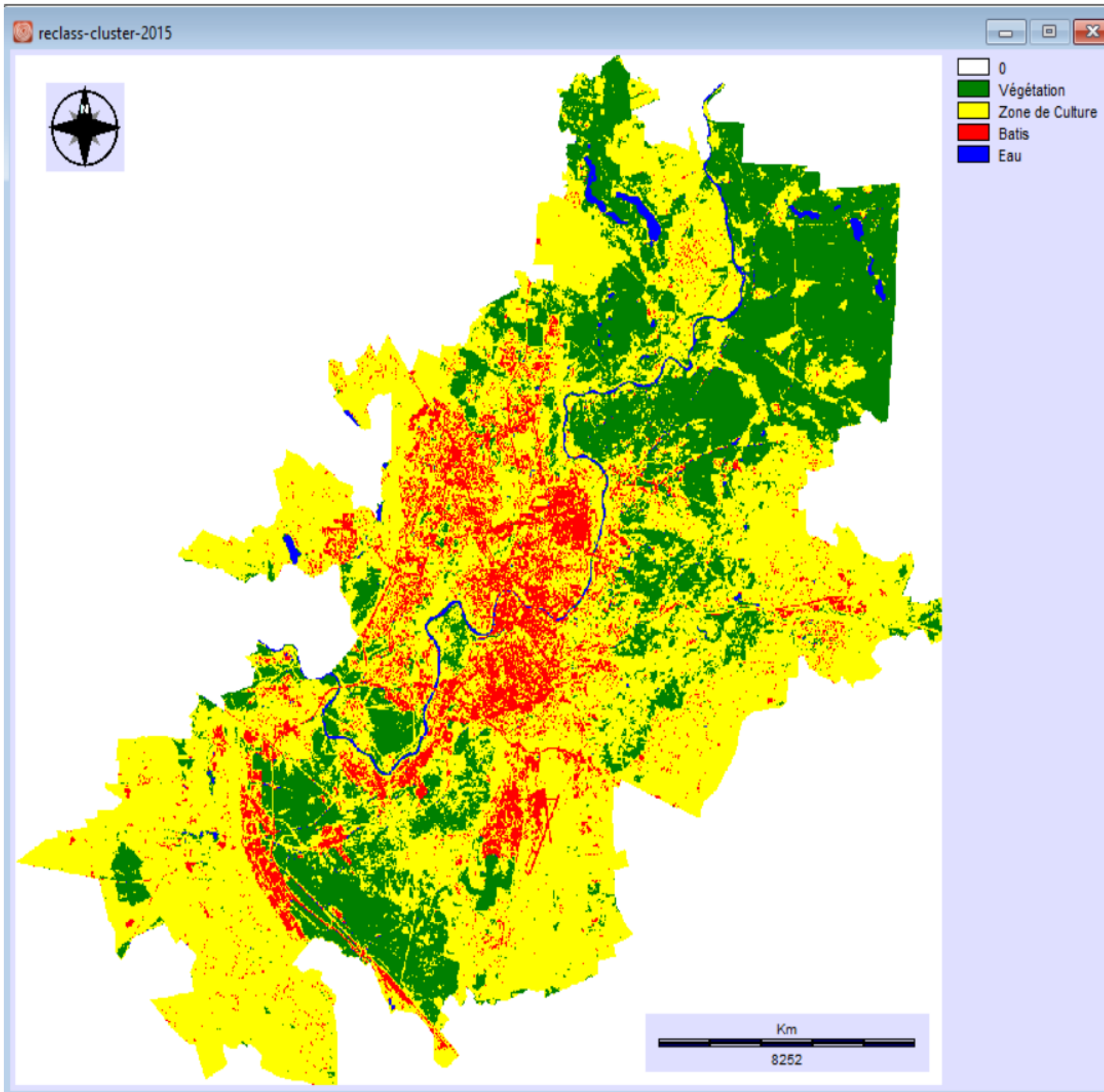
NDWI-2024



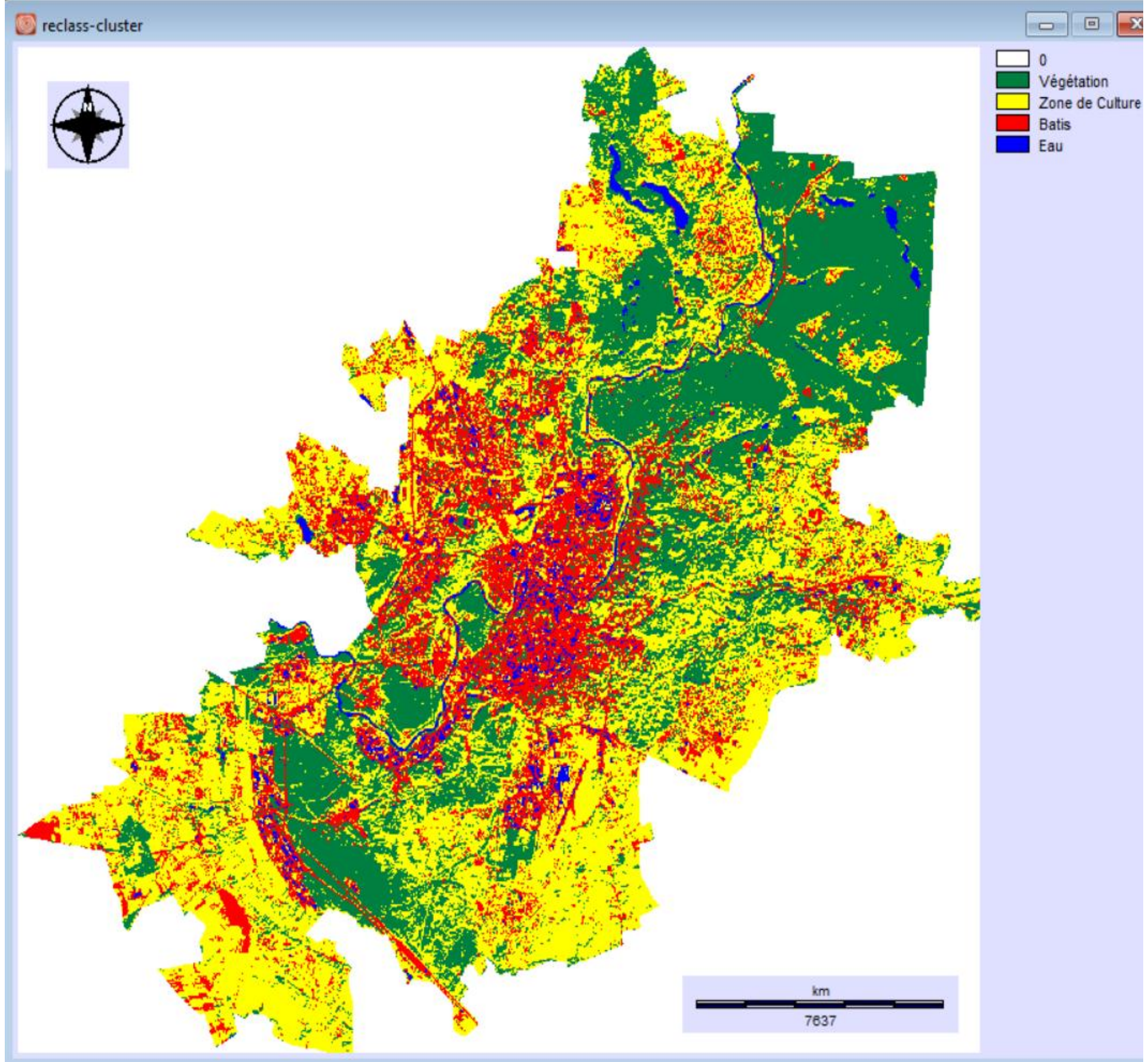
Composite-NDWI-2015-2024-B2



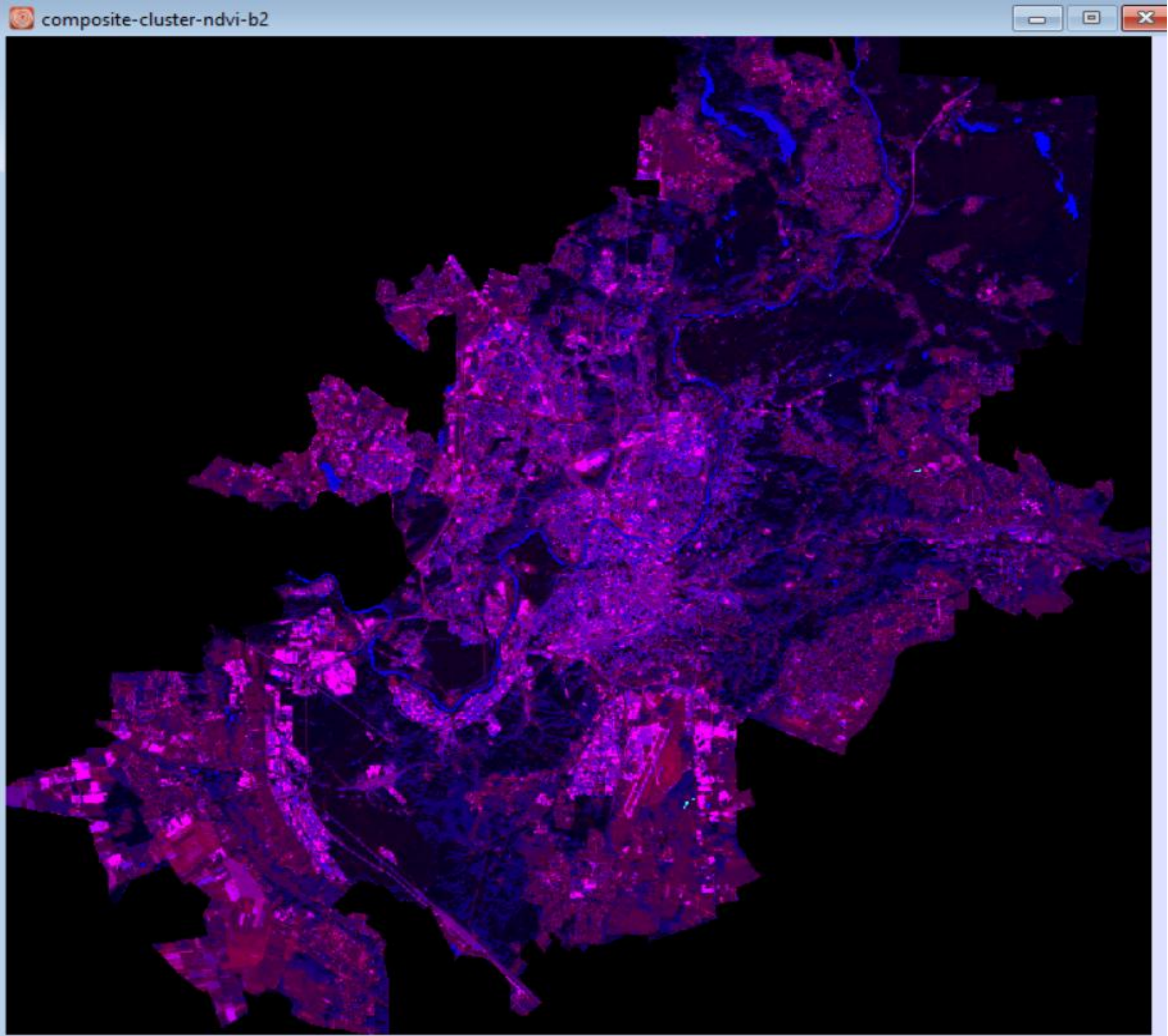
Land use 2015



Land use 2024

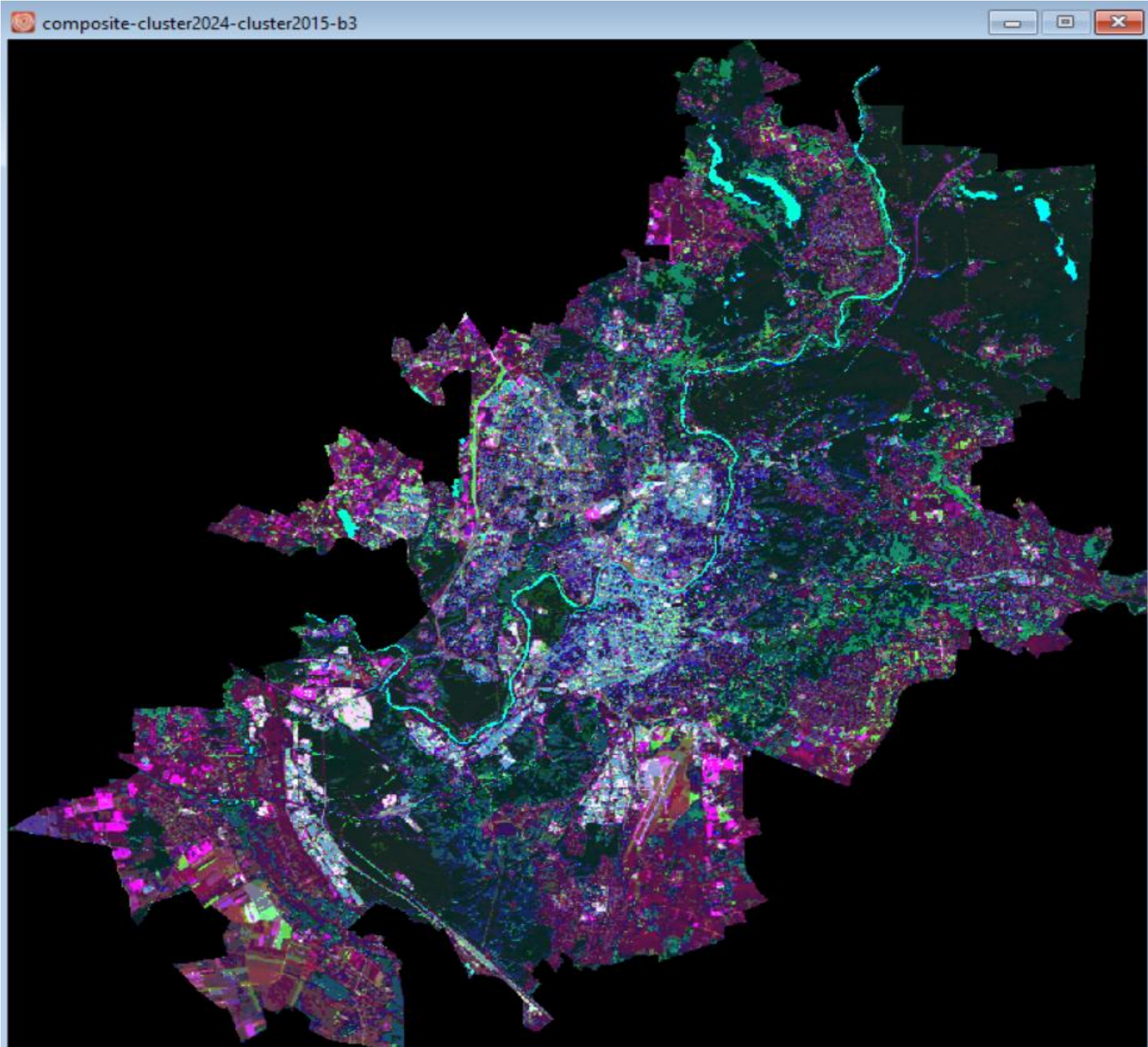


Composite-Cluster2024-NDVI-B2



The combination of Cluster 2024, NDVI 2024 and Band 2 allows us to clearly see the wettest areas of the city of Vilnius. We can have a strong presence of water to the north of the city and a stream that runs through the center of the city.

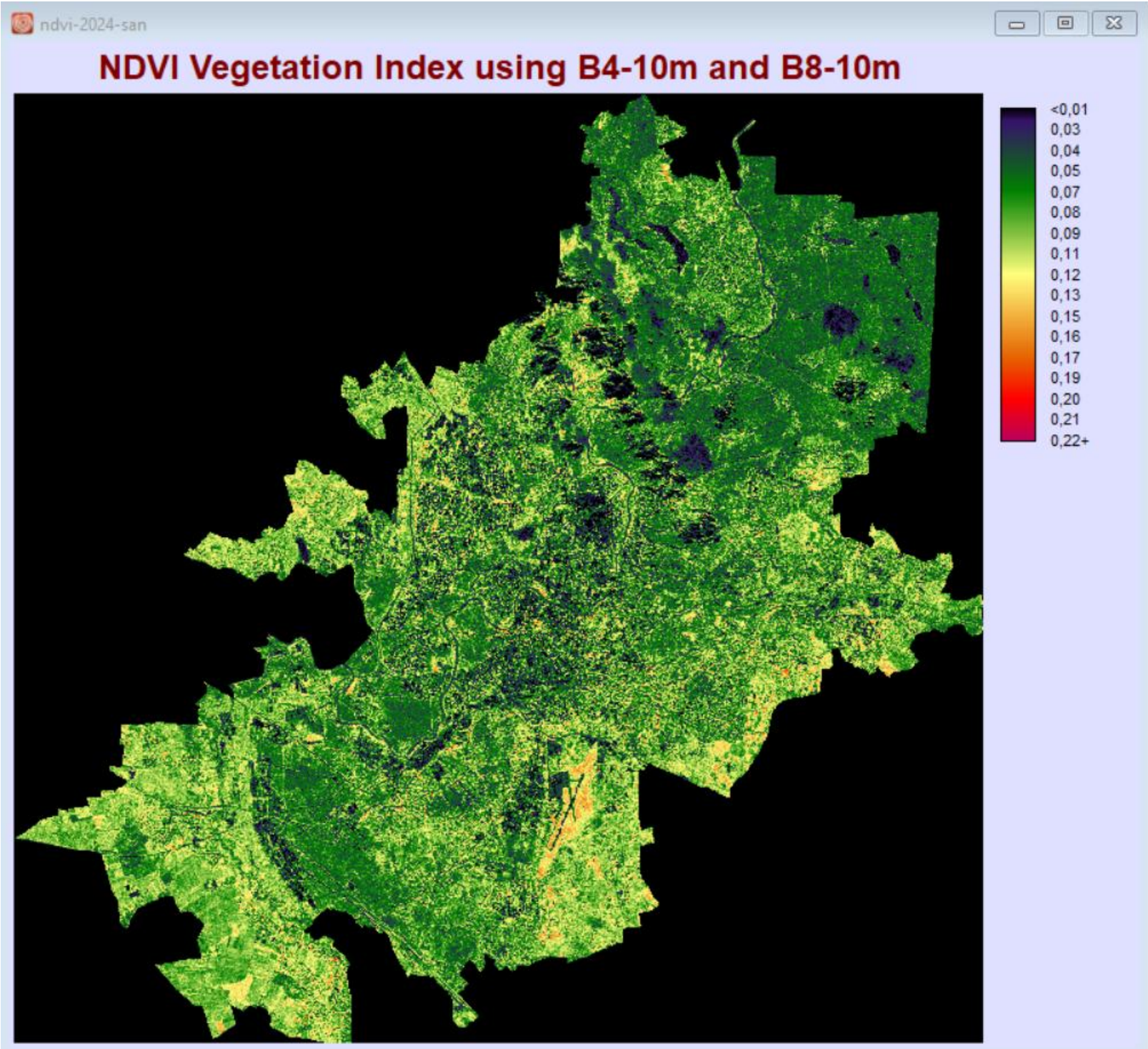
Composite-Cluster2024-Cluster2015-B3



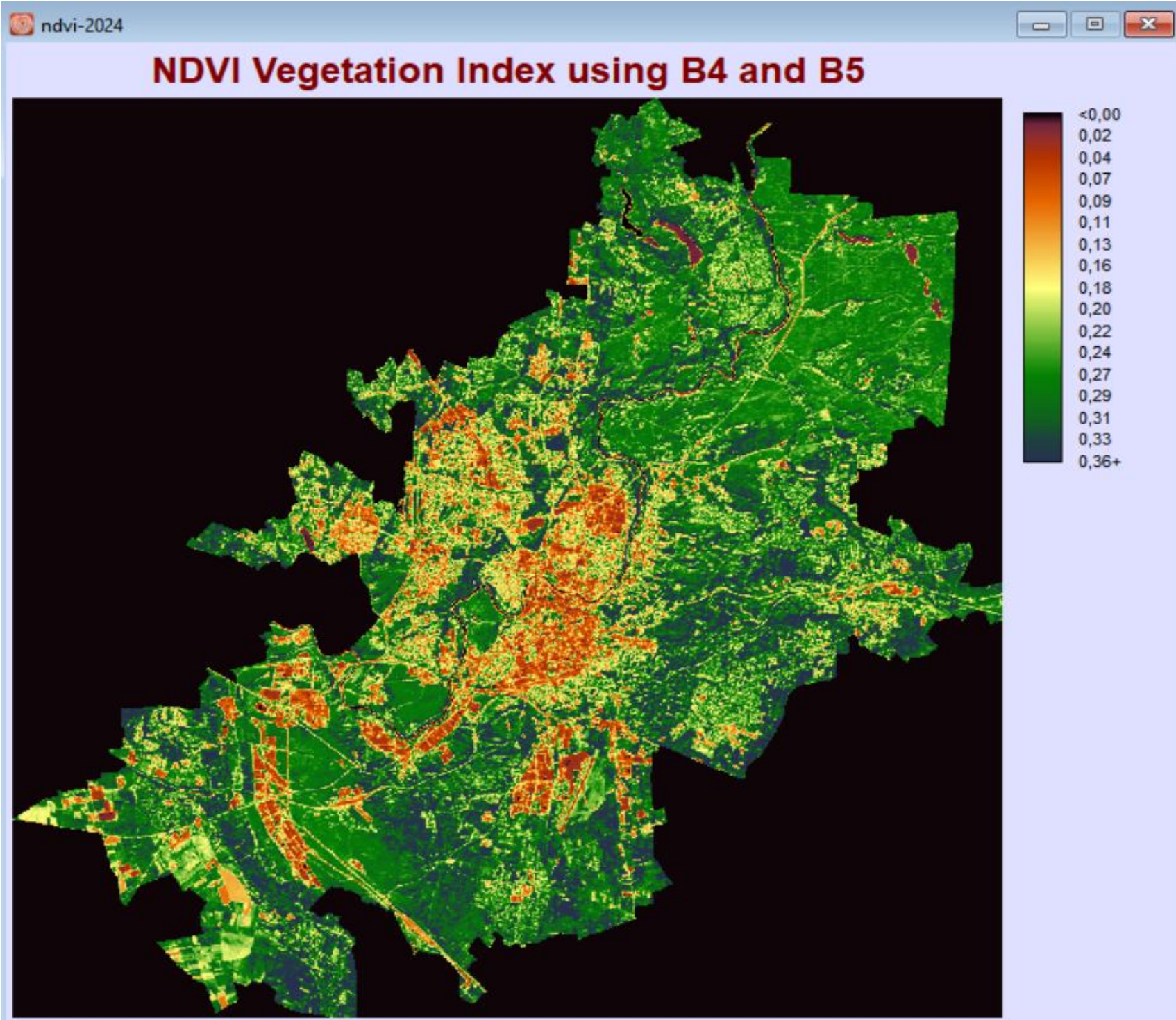
The combination of Cluster 2024, Cluster 2015 and Band 3 allows us to clearly see the vegetation and growing areas that are more present in the south of the city.

LANDSAT 8 2024 AND SENTINEL 2 2024 IMAGERY COMPARISON

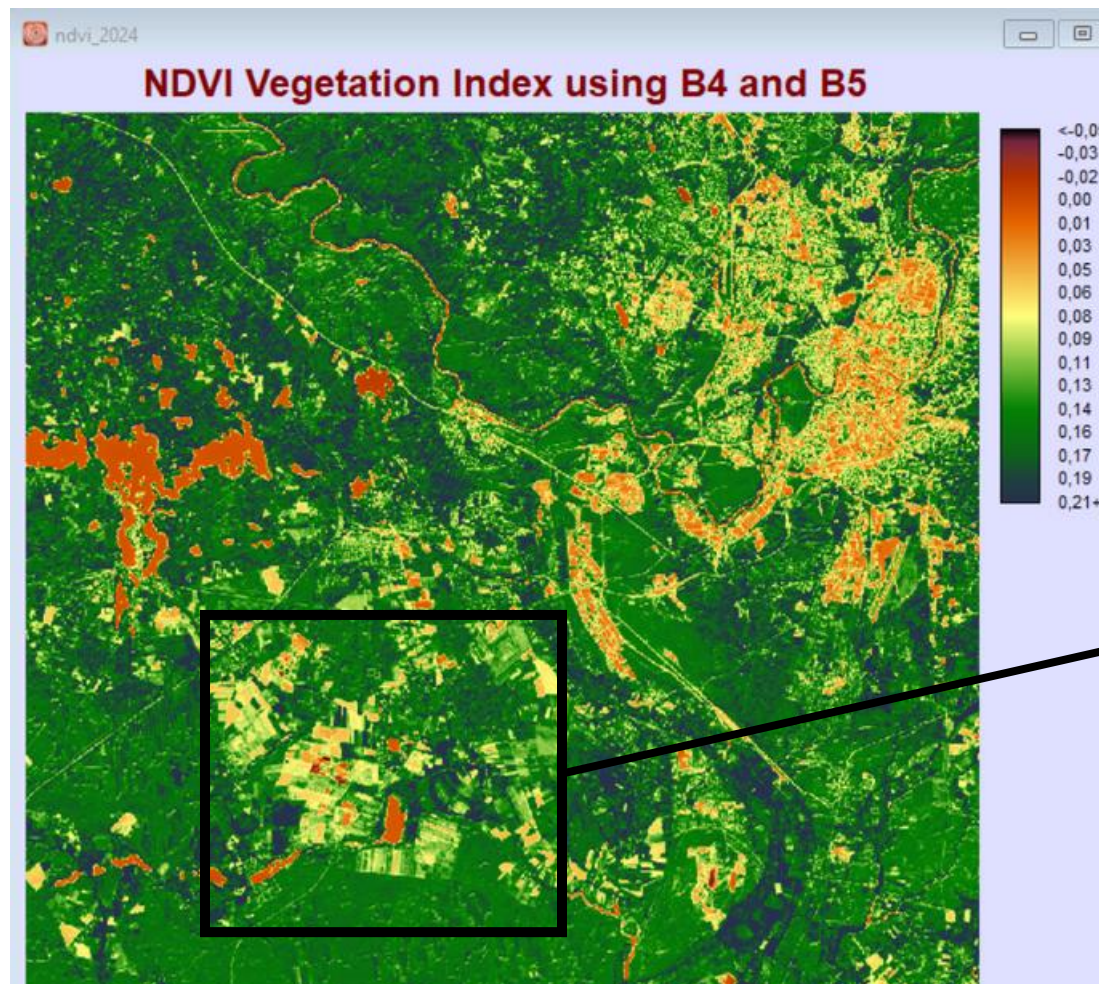
Sentinel 2



Landsat 8



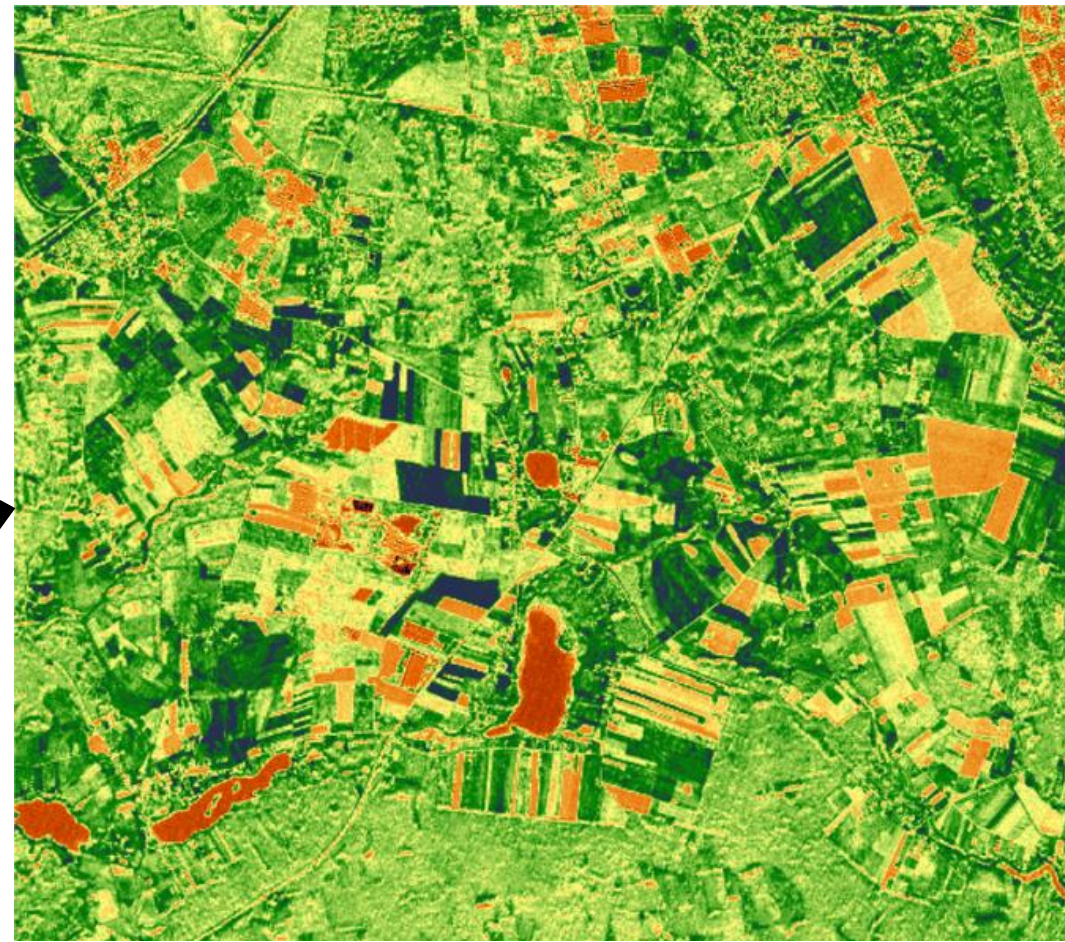
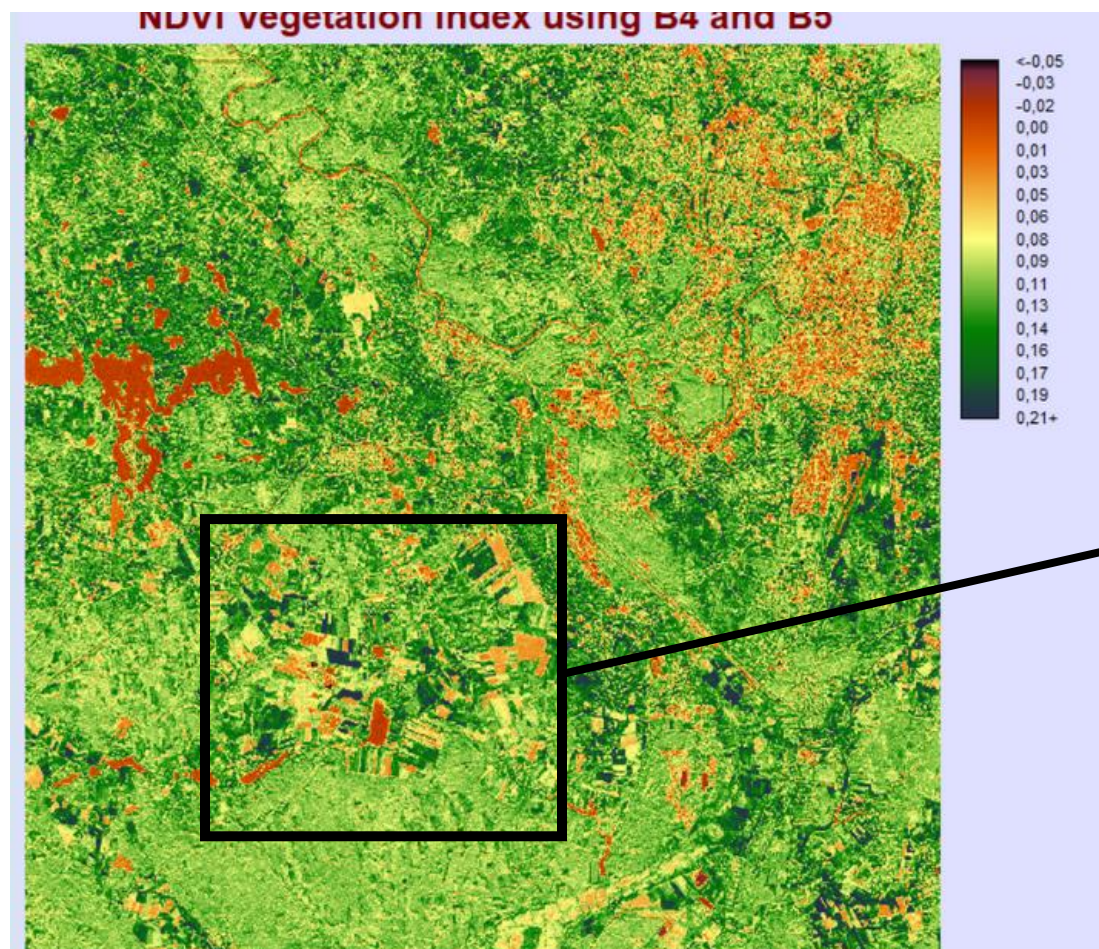
LANDSAT 8-2024



Comparing the Landsat 8 image with the Sentinel 2 image, we see that:

The Landsat 8 image shows us that a large part of the vegetation is healthy, but by account the Sentinel 2 image shows us that the vegetation in this area is not healthy because due to urbanization and the increase in cultivation areas, it is starting to shrink.

SENTINEL 2-2024



DISCUSSION

- ❖ Sustainable urban agriculture is an innovative response to the challenges posed by rapid urbanization and environmental issues.
- ❖ Geospatial data (Landsat 8, Sentinel 2) and indices (NDVI, EVI, NDWI) have made it possible to identify:
 - Densely vegetated areas that are conducive to urban agriculture.
 - Wetlands, such as those north of Vilnius, have specific potential.
- ❖ Key Results:
 - A significant increase in the area of vegetation cover in some southern parts of Vilnius between 2015 and 2024.
 - Clear visualization of trends thanks to advanced processing on Terrset and QGIS.
- ❖ Limitations identified:
 - Influence of atmospheric conditions on certain indices such as NDVI.
 - Need to integrate socio-economic data to complement geospatial analysis.
- ☐ Prospects:
 - Incorporate human factors (community preferences, available infrastructure).
 - Expand the study to assess the long-term impacts of the

CONCLUSION

- ❖ Geospatial modelling tools offer an effective method for planning sustainable urban agriculture.
- ❖ The results obtained for Vilnius show that:
 - Strategic areas can be identified to maximize the sustainability of urban food systems.
 - The combined indices (NDVI, EVI, NDWI) allow a detailed analysis of environmental dynamics.
- ❖ Practical implications:
 - This data can guide local policies to promote urban agriculture.
 - Vilnius can serve as a model for other cities in the implementation of sustainable agriculture.
- ❖ Recommendations:
 - Adopt a multidisciplinary approach integrating environmental, social and economic dimensions.
 - To further explore the interactions between urban development and sustainable agriculture.